

Multiple Choice

1. **Answer:** C

Options: A) Cancer cells have no telomere strands; B) Normal cells become cancerous when their telomere strands are used up; C) Cancer cells have no upper limit on cell division; D) Normal cells are protected from cancer when they have telomerase

Note: Cancer cells can bypass normal regulatory mechanisms that limit cell division, allowing them to proliferate indefinitely, unlike normal cells which have a finite number of divisions before undergoing senescence.

2. **Answer:** A

Options: A) Gary who is still married; B) Leo who never married; C) Josh who is divorced; D) Caleb who is remarried

Note: Gary is likely to be the happiest in retirement because studies have shown that individuals who maintain strong social connections, such as marriage, tend to experience greater emotional well-being and life satisfaction.

3. **Answer:** B

Options: A) Traits; B) Personal concerns; C) Coping strategies; D) Identity

Note: Personal concerns, which encompass an individual's goals, values, and coping strategies, tend to evolve significantly over time in response to life experiences and developmental stages, making them the level of personality description that shows the most change with age.

4. **Answer:** D

Options: A) Circular ds DNA; B) Small linear ss DNA; C) ds DNA genome; D) RNA genome with RT

Note: The correct answer is based on the fact that parvoviruses possess a single-stranded RNA genome that utilizes reverse transcription (RT) during their replication process, distinguishing them from many other viruses.

5. **Answer:** B

Options: A) Influenza; B) MMR; C) Yellow Fever; D) Rabies

Note: The MMR vaccine, which protects against measles, mumps, and rubella, is considered the most successful and widely used travel vaccine for international visitors due to the global prevalence of these diseases and the significant risk they pose to unvaccinated travelers.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: The statement is false because DNA damage and repair theory attributes senescence primarily to cumulative DNA damage from various sources, including environmental factors and normal metabolic processes, rather than radiation from antioxidants, which are actually known to mitigate oxidative stress rather than cause it.

7. **Answer:** True

Options: A) True; B) False

Note: The prepuce, or foreskin, is the fold of skin that covers the glans of the penis, and it is surgically removed during male circumcision, making the statement true.

8. **Answer:** True

Options: A) True; B) False

Note: In X-linked dominant inheritance, males have only one X chromosome to pass on, which means they can only transmit the trait to their daughters, who receive their father's X chromosome, while their sons receive the Y chromosome from their father, not the X.

9. **Answer:** False

Options: A) True; B) False

Note: Adaptive immunity is characterized by a slower response to foreign proteins due to its reliance on the activation and clonal expansion of specific lymphocytes, unlike the innate immune response, which acts immediately and nonspecifically.

10. **Answer:** True

Options: A) True; B) False

Note: Public health ethics prioritizes the health of communities and populations as a whole, addressing collective health issues and promoting policies that benefit the greater good rather than focusing solely on individual patient care.

Long Form Response

11. **Answer:** The prone position, where a patient lies flat on their stomach, has significant implications and applications in patient care, particularly in the management of respiratory conditions. One of the primary benefits of this position is its ability to improve oxygenation and lung function by promoting better ventilation-perfusion matching in the lungs, especially in patients with acute respiratory distress syndrome (ARDS). However, there are associated risks, such as pressure injuries and complications in patients with certain conditions, like spinal injuries or unstable fractures. The prone position is commonly utilized in critical care settings and during certain surgical procedures to enhance patient outcomes. Ultimately, while the prone position can be beneficial in specific scenarios, careful consideration of patient safety and comfort is essential.

Note: correct because it accurately identifies the prone position's role in enhancing respiratory function and oxygenation, particularly in ARDS patients, while also acknowledging the potential risks involved, thus providing a balanced overview of its implications and applications in health sciences.

12. **Answer:** Next generation sequencing (NGS) has several limitations in health sciences that can affect its applicability and effectiveness. For instance, NGS may not be suitable for detecting structural variants or large genomic rearrangements, which require different methodologies such as optical mapping or traditional karyotyping. Additionally, in cases where there is low tumor mutational burden, such as some forms of cancer, NGS may not provide sufficient actionable information, potentially leading to missed therapeutic opportunities. Furthermore, the high cost and complexity of data interpretation can limit its use in resource-limited settings or for routine clinical assessments. These scenarios illustrate that while NGS is a powerful tool, it is not universally applicable and should be considered alongside other diagnostic methods. *Note:* correct because it identifies specific limitations of next generation sequencing, such as its ineffectiveness in detecting structural variants and its potential insufficiency in providing actionable information for cancers with low mutational burden, thereby illustrating scenarios where NGS may not be applicable.